

Deep Learning

Assignment-03 (Report)

Course Name: Deep Learning

Course Code: CS508

Session: Spring 2026

Instructor: Dr. Mohsen Ali

Submitted by:

Mubashar Hussain

MSDS24031

Date of Submission: March 27, 2026

Contents

1	Model Architecture	2
2	Dataset Sampling Strategy	2
2.1	Contrastive Sampling	2
2.2	Triplet Sampling	2
2.3	Dataset Preparation	2
3	Loss Functions	2
3.1	Contrastive Loss	2
3.2	Triplet Loss	2
4	Training Setup and Hyperparameters	2
4.1	Trainin Setup Justification	3
5	Training Behavior (Loss Curves)	3
6	t-SNE Visualization	5
7	Retrieval Performance	6
8	Embedding Analysis	6
8.1	Same Class Similarity	6
8.2	Different Class Similarity	6
9	Retrieval Visualizations	7
10	Discussion	9
11	AI Usage for learning and guidance	9

1 Model Architecture

I implemented deep metric learning model using ResNet-50 as backbone, where pretrained ResNet-50 (ImageNet) was used to extract features from images, final classification layer was removed & replaced with fully connected projection layer that maps features to 128-dimensional embedding space, then L2 normalization was applied so all embeddings lie on the unit hypersphere, and this normalization helps making distance comparisons more meaningful & stable during training and retrieval. I froze backbone, and resized image size to 128×128 to reduce computational cost & **prevent dangerous overheating**, only projection head was trained.

2 Dataset Sampling Strategy

2.1 Contrastive Sampling

Pairs of images were created: where positive pair means same class, and negative pair means different class. Both types were sampled with the equal probability.

2.2 Triplet Sampling

Triplets were formed as: where anchor is reference image, positive is same class as anchor, negative is different class. For hard negative mining, hardest positive & hardest negative samples were selected within the each batch.

2.3 Dataset Preparation

Because of my computational limitations, I used subset of 5000 images from Caltech-101 dataset, where subset was fixed using precomputed indices to ensure the consistency. Dataset was split into: 70% training, 15% validation, 15% test. I used fixed random seed (42) to ensure reproducibility of splits.

3 Loss Functions

3.1 Contrastive Loss

Contrastive loss minimizes distance between similar pairs & increases distance between dissimilar pairs.

3.2 Triplet Loss

Triplet loss ensures that anchor is closer to positive than negative by margin, as triplet loss directly enforces relative distance, so it is more suitable for retrieval tasks.

4 Training Setup and Hyperparameters

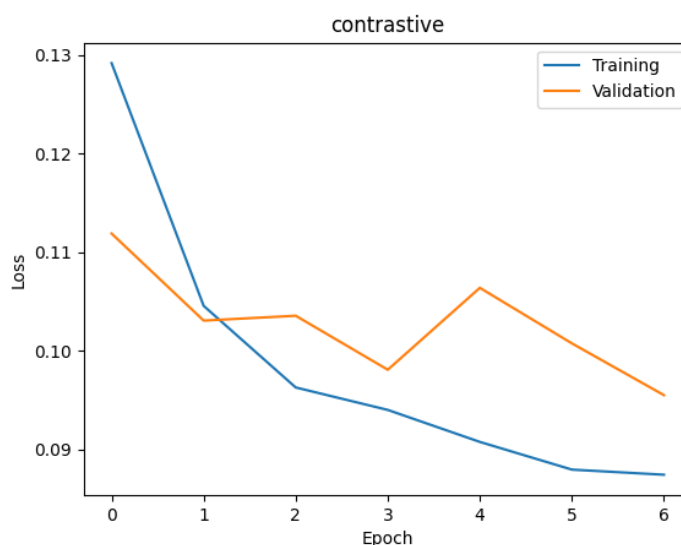
I used batch size of 8 due to my hardware limitations. Moreover to reduce computational cost & **prevent dangerous overheating**, I froze backbone and resized input images to

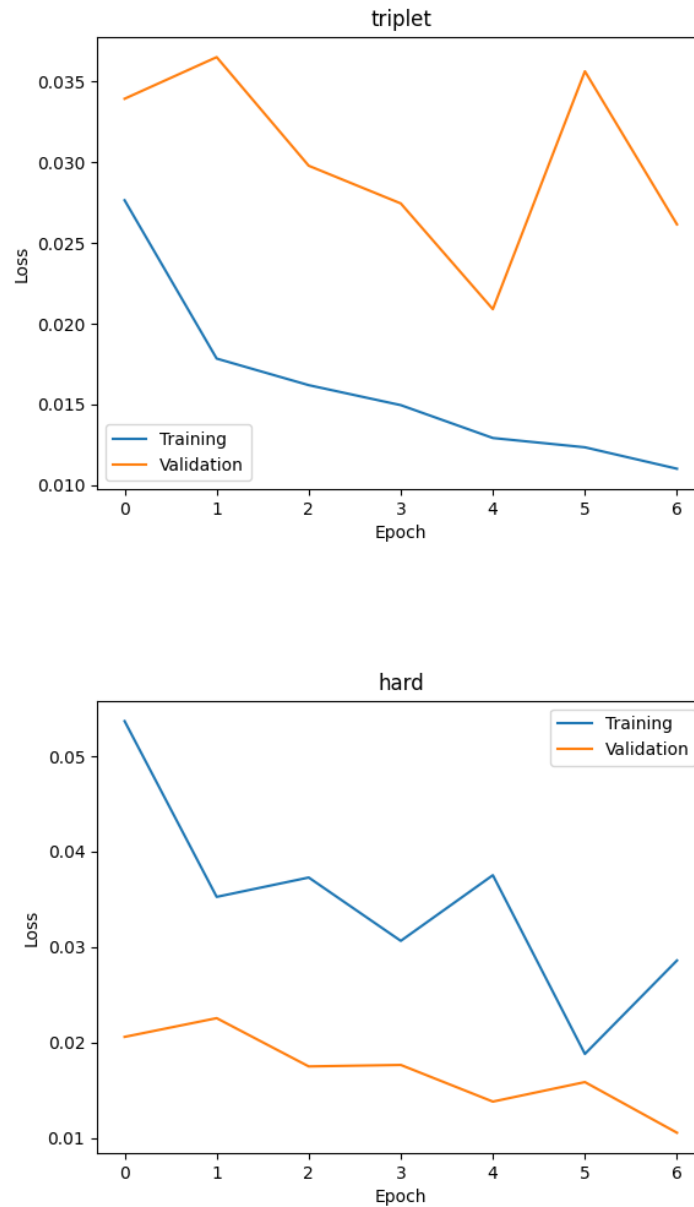
128×128. Adam optimizer was used with lr (1e-3). Model trained for 7 epochs (**which was actually sufficient for convergence without overfitting**), where applied early stopping based on validation loss to prevent overfitting and saving best model. All experiments were conducted on CPU.

4.1 Trainin Setup Justification

First I tried to check training behaviour with unfreezed backbone and image size of 224×224, but **my laptop was overheated to dangerous extent in early few minutes of training**, so I decided to freeze backbone to reduce computation and also prevent large gradient updates from disturbing pretrained features, and this actually helped stabilize training on my limited hardware. Moreover I decided to reduce input image size to 128×128 to decrease computational cost, while preserving important visual features. I used subset of 5000 images to speed up training, while still maintaining diversity across the classes.

5 Training Behavior (Loss Curves)

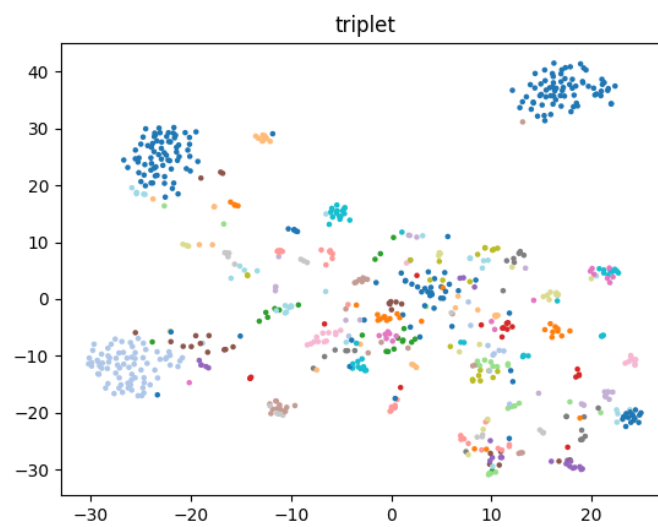
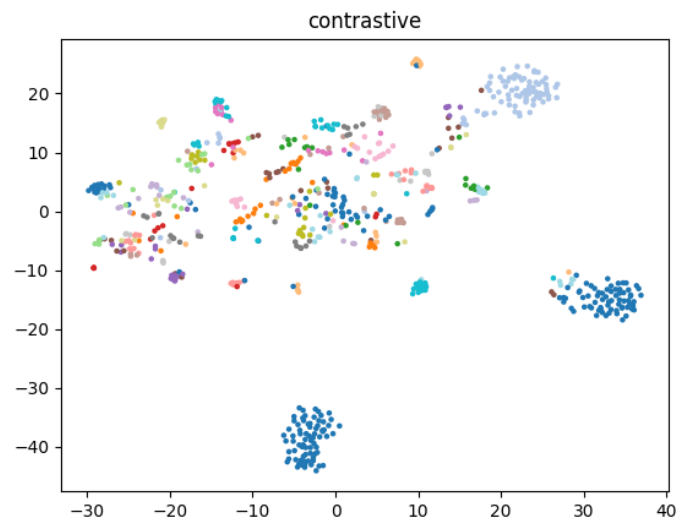


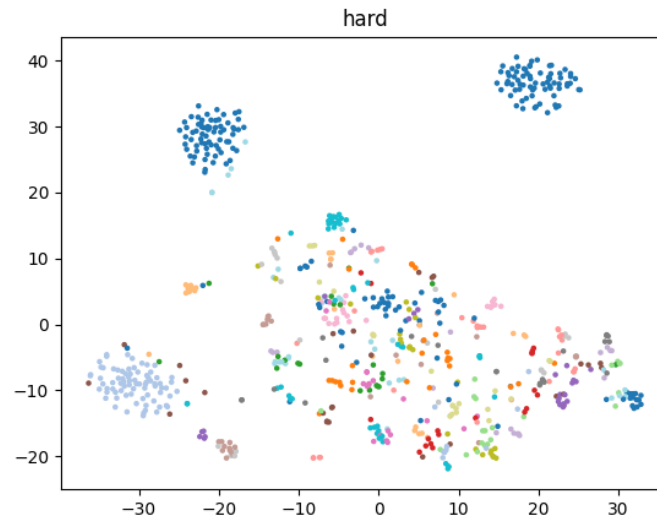


Here contrastive-loss is shown stable & smooth training, triplet (random) also decreased steadily but the validation fluctuated, however hard-mining had unstable training loss, but very low validation loss. Despite low validation loss, hard-mining did not perform well during retrieval.

Results show that triplet-loss with random sampling provides best balance between intra class compactness & inter class separation, and triplet-loss with random sampling achieved best performance, but contrastive-loss performed reasonably well but was slightly lower, while hard-negative-mining gave lowest Recall values. This shows that lower loss does not always mean better retrieval performance.

6 t-SNE Visualization





Here contrastive embeddings is showing partial clustering, but many points were mixed, and triplet embeddings showing better separation & more meaningful clusters, however hard-mining created some tight clusters, but overall structure was not organized well.

7 Retrieval Performance

Models were evaluated using Recall@K.

Method	Recall@1	Recall@5
Contrastive	0.700	0.871
Triplet (Random)	0.747	0.896
Triplet (Hard)	0.688	0.848

8 Embedding Analysis

8.1 Same Class Similarity

Different images of same class produced cosine scores: contrastive (0.847), Triplet (0.767), hard (0.785). Clearly contrastive loss produced highest similarity for same class images, which means that it created very compact clusters.

8.2 Different Class Similarity

Different class cosine scores were: contrastive (0.446), triplet (-0.029), hard (0.579). Here triplet-loss produced lowest similarity for different classes, which is desirable, so it shows contrastive-loss focuses more on pulling similar samples together, while the triplet-loss better separates different classes.

9 Retrieval Visualizations



Figure 1: Contrastive Retrieval



Figure 2: Contrastive Retrieval



Figure 3: Contrastive Retrieval

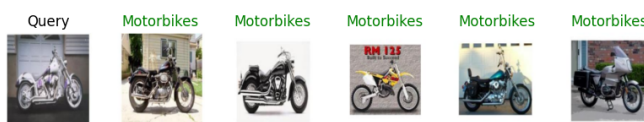


Figure 4: Contrastive Retrieval



Figure 5: Triplet Retrieval

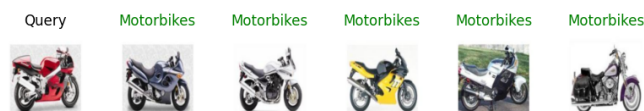


Figure 6: Triplet Retrieval



Figure 7: Triplet Retrieval



Figure 8: Hard Retrieval



Figure 9: Hard Retrieval



Figure 10: Hard Retrieval

These retrieval results showing model is able to correctly group visually similar images, for many queries such as airplanes, umbrellas, laptops, and motorbikes, all of the top5 retrieved images belong to same class, indicating strong feature learning. While in some case, few incorrect results appear, and these errors occur due to visual similarity between the classes, but overall model performing well in capturing the global structure & semantic similarity with most of retrieved results are correct.

10 Discussion

It can be seen in all experiments that triplet-loss with random sampling gave best overall performance, although the contrastive-loss produces very high similarity for same class images, it does not separate different classes effectively. Triplet loss balances both objectives it brings similar images closer, and pushes different images apart. Hard negative mining focuses on most difficult samples within batch, while this reduces loss, it can sometimes lead to unstable training & poor global structure of embedding space. So overall, the good retrieval depends more on relative distances rather than the absolute similarity.

11 AI Usage for learning and guidance

<https://chatgpt.com/share/69c68eb3-75c8-8323-853e-c1827d7eb080>

<https://chatgpt.com/share/69c68ffa-2b20-8321-b4eb-aba067a1617c>

<https://chatgpt.com/share/69c69018-3060-8321-bf5c-d784662fd4bd>